

PROJECT PROPOSAL

TripForge: A Collaborative Trip Planning Web Application

1. Project Title

TripForge: A Collaborative Trip Planning Web Application

2. Project Overview

TripForge is a web application that lets a group of friends or students plan an entire trip in one shared workspace. Today, groups coordinate through Messenger or WhatsApp, write itineraries in Notes, and track costs in a calculator or Excel. TripForge brings these into one platform.

A user creates an account, creates a trip (name, dates, budget), and invites friends. Together they build the itinerary, view locations on a map, check weather forecasts, suggest and vote on ideas, run polls, split expenses, discuss plans, and prepare a packing checklist. Everything ends in a single Final Trip Summary.

Unlike a traditional management system with one admin, TripForge is built around collaboration: every member takes part in planning.

3. Problem Statement

Group trip planning is usually scattered and disorganized:

- Information is spread across chat apps, notes, and spreadsheets, so details get lost or duplicated.
- Group decisions (destination, activities, restaurants) are slow and often dominated by a few voices in chat threads.
- Expenses paid by different people are hard to track, which causes confusion about who owes what.
- Planning is often done without checking route distances or weather, which affects safety and enjoyment.
- There is no single, final, shareable view of the whole plan.

There is a need for one centralized platform where a group can plan together, decide democratically, and keep all trip information organized.

4. Objectives

1. Provide a single shared workspace for all aspects of trip planning.
2. Enable real collaboration, where all members can contribute, edit, and vote.
3. Simplify group decision-making through idea voting and polls.
4. Integrate Google Maps so locations, distances, and routes are visible.
5. Integrate a Weather API so groups can plan according to forecasts.
6. Automate expense calculation and per-member share tracking.
7. Keep members informed through in-app notifications.
8. Generate a complete Final Trip Summary in one place.
9. Build the system with a secure, maintainable stack (React, Node.js/Express, MySQL, JWT).

5. Target Users

- **University students** planning tours and group trips.
- **Friend groups** organizing weekend getaways or vacations.
- **Small groups** such as families, clubs, or office teams planning outings.
- **Trip creators/organizers**, who create trips and invite members.
- **Trip members**, who contribute ideas, vote, add expenses, and update plans.

6. Project Scope

In scope

- User registration, login, and secure authentication
- Trip creation and member invitations
- Shared workspace with itinerary, ideas/voting, polls, expenses, discussion, packing checklist, and notifications
- Google Maps and Weather API integration
- Expense calculation and share tracking
- Final Trip Summary

Out of scope

- Real payment processing (expenses are calculated and tracked only)
- Real-time chat (replaced by a database-backed Discussion/Comments section)
- Booking of transport, hotels, or tickets
- Push, email, or SMS notifications (only in-app notifications)
- Native mobile applications (a responsive web app is the target)

7. Functional Requirements

ID	Module	Requirement
FR1	Authentication	Users can register, log in, and log out securely.
FR2	Trip Management	Users can create a trip with name, dates, and budget, and edit or delete it.
FR3	Invitations	The trip creator can invite friends to join a trip.
FR4	Itinerary	Members can add, edit, delete, and reorder day-wise activities with times.
FR5	Maps	Users can add activity locations and view them on a map, with distance and route between locations.
FR6	Weather	Users can view temperature, conditions, rain information, and forecast for the destination and trip dates.
FR7	Ideas & Voting	Members can post ideas, vote on them, and add approved ideas to the itinerary.
FR8	Polls	The trip creator can create polls with multiple options; members vote and results are shown automatically.
FR9	Expenses	Members can add expenses (title, amount, payer). The system calculates the total and each member's share.
FR10	Discussion	Members can post comments and replies, which are saved and viewable later.

ID	Module	Requirement
FR11	Packing Checklist	Members can add items, mark them packed, and assign items to members.
FR12	Notifications	Members receive in-app notifications for new polls, suggestions, expenses, and itinerary changes.
FR13	Trip Summary	The system displays a full summary: dates, destination, members, map locations, weather, itinerary, expenses, shares, checklist, and final decisions.

8. Non-Functional Requirements

- **Security:** Passwords hashed with bcrypt; sessions secured with JWT; only trip members can access a trip's data.
- **Usability:** Clean, intuitive interface that new users can learn without training.
- **Performance:** Pages load quickly, and API responses should typically complete within a few seconds.
- **Reliability:** Data is persistently stored in MySQL, with consistent and accurate expense calculations.
- **Scalability:** A modular architecture so features can be extended later.
- **Compatibility and responsiveness:** Works on modern browsers and adapts to desktop, tablet, and mobile screens.
- **Maintainability:** Separation of frontend, backend, and database layers with clean, documented code.
- **Availability of integrations:** Graceful handling of Google Maps or Weather API failures, such as error messages instead of crashes.

9. Key Features

1. Shared Trip Workspace: one hub for all trip information
2. Collaborative Itinerary: day-wise plans that every member can add, edit, and reorder
3. Google Maps Integration: locations, distances, and routes
4. Weather Information: temperature, conditions, rain, and forecast for trip dates
5. Trip Ideas & Voting: democratic selection of activities
6. Poll System: quick group decisions such as restaurant choice
7. Expense Sharing: automatic totals and per-member shares
8. Trip Discussion: persistent comments and replies without real-time complexity
9. Packing Checklist: items with member assignments
10. In-App Notifications: updates on group activity
11. Final Trip Summary: the complete plan in one view

10. Technology Stack

Layer	Technology
Frontend	React.js
Backend	Node.js with Express.js
Database	MySQL
Authentication	JWT and bcrypt

Layer	Technology
Maps	Google Maps API
Weather	Weather API

Workflow: Account creation → Trip creation → Friend invitations → Location selection → Weather check → Ideas → Voting → Itinerary → Expense calculation → Discussion → Packing checklist → Final Trip Summary.

11. Expected Outcome

On completion, TripForge will be a working web application where a group can:

- Plan an entire trip collaboratively on one platform instead of juggling several apps
- Make fair, transparent decisions through voting and polls
- Visualize destinations and routes and plan around weather conditions
- Track shared expenses and know exactly what each member owes
- Stay updated through notifications and keep all discussions saved
- Review one final summary containing the entire trip plan

The project will demonstrate full-stack development (React, Node.js/Express, MySQL), secure authentication, third-party API integration, and a collaboration-focused design that solves a real everyday problem for students and friend groups.